

Speech Is Not Belief: Social Fidelity Decay in LLM Agent Societies

Anonymous submission

Abstract

Generative-agent work shows that information can spread through a simulated society, but not whether the spread is faithful. We introduce a social fidelity probe that injects a single ground-truthed update into an LLM-agent society, lets the agents relay it through rounds of pairwise conversation, and then interviews every agent to measure what the society holds. Across capability, connectivity, memory architecture, and a persistent authoritative source, natural levers fail to restore held truth; only directly overwriting every agent’s memory does. The sharpest result is a dissociation: an authoritative source makes agents increasingly say the current value while the society’s held belief stays near baseline—speech moves without belief. We trace the mechanism to path-dependent entrenchment: the version established early and broadly wins, and late or narrow correction can be uttered without being adopted. The effect survives a long horizon (truth peaks early then decays), three scenarios, thicker personas, and a semantic-judge validation of the metric.

Introduction

Large language model (LLM) agents increasingly form societies that talk, remember, and coordinate (Park et al. 2023; Li et al. 2023a; Wu et al. 2023; Chen et al. 2023). A celebrated demonstration is that information spreads through such a society, and Park et al. (2024) show that a single agent can be made remarkably faithful to a real person. Between “information spreads” and “an agent is individually faithful” lies a question neither line answers: when a society passes a changing fact through conversation, does it come to *hold* the truth, or merely *repeat* it?

This is the distinction between *reach* and *fidelity*, long drawn in misinformation research (Lazer et al. 2018; Vosoughi, Roy, and Aral 2018). It matters now because agent societies are themselves used as instruments—to simulate opinion, deliberate, or aggregate judgments—and their collective outputs are read as if they were the society’s beliefs. If a society can be made to say something it does not hold, such readings are unsafe. Individual fidelity does not imply collective fidelity.

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We operationalize this question with a controlled social fidelity probe (Figure 1). We inject a single ground-truthed update at one source, let $N=25$ agents relay it through rounds of pairwise conversation using generative-agent reflection memory, and then interview every agent to measure what the society holds. We separate what agents say in meetings from what they hold at interview, and we test the natural repair levers—model capability, connectivity, memory architecture, and a persistent authoritative source—against a positive control that overwrites every agent.

The main result is a speech–belief dissociation (Figure 2): a persistent authoritative source makes agents increasingly say the current value while what the society holds stays near baseline. Speech moves without belief. The dissociation sits inside a broader decay process: truth rises early and then erodes to a low, unknown-dominated steady state (Figure 3). Natural repair levers do not close the gap—capability, connectivity, memory architecture, and authority all fall far short of a direct all-agent broadcast (Figure 4). The mechanism is path-dependent entrenchment, not recency (Figure 5): an early, broad correction succeeds where a late one fails. The pattern replicates across three scenarios and under richer personas (Figure 6).

Our contributions are:

- a **social fidelity probe** that separates what agents hear, what they say, and what they later hold after a ground-truthed update;
- a robust **speech–belief dissociation**: authority can move utterances toward truth while held belief remains near baseline; and
- an empirical and mechanistic account of **social fidelity decay**: natural repair levers fail, while early and broad entrenchment explains when truth persists.

Related Work

Agent societies and communicative agents. Park et al. (2023) establish a setting in which LLM agents talk, remember, and coordinate in a believable social world, and communicative-agent systems make agent-to-agent dialogue a standard substrate (Li et al. 2023a; Wu et al. 2023; Chen et al. 2023; Hong et al. 2024; Qian et al. 2024; Gao et al. 2024). Interactive-agent methods further interleave reasoning, action, and observable traces (Yao et al. 2023), while

social-evaluation environments broaden assessment beyond isolated QA (Zhou et al. 2024). Multi-agent collaboration work studies theory-of-mind and task-state failures (Li et al. 2023b), and recent work shows a single agent can be made faithful to a real individual (Park et al. 2024). We ask a narrower reliability question: whether a changing fact remains current after social transmission.

Reach versus fidelity in (mis)information. Misinformation research separates how far content spreads from whether it is believed and acted upon (Lazer et al. 2018; Vosoughi, Roy, and Aral 2018; Friggeri et al. 2014). With LLM agents, Maurya et al. (2025) reproduce misinformation propagation through social networks, but measure propagation or vulnerability. Our probe shifts the endpoint from reach to held-belief fidelity after a ground-truthed change.

Factuality, debate, and the speech-belief gap. Factuality evaluation treats truth as a semantic behavioral target rather than fluency (Lin, Hilton, and Evans 2022; Kadavath et al. 2022; Manakul, Liusie, and Gales 2023; Farquhar et al. 2024; Li et al. 2023c; Niu et al. 2024). Calibration and internal-state work further caution against reading surface text as the whole epistemic state (Azaria and Mitchell 2023; Tian et al. 2023). Multi-agent debate is the optimistic counterpoint—structured discussion can improve judged answers (Du et al. 2024; Michael et al. 2023; Liang et al. 2024; Khan et al. 2024)—yet explicit correction can fail without reliable grounding (Huang et al. 2024). We identify a complementary society-level failure: the correct statement can win space in conversation without becoming the answer agents later hold.

Memory, communication, and recursive degradation. Agent-memory and reflection methods show that durable state must be engineered—exposure in context is not the same as retrievable belief (Shinn et al. 2023; Zhong et al. 2024; Packer et al. 2024; Wang et al. 2023; Xu et al. 2025). Human communication work provides parallel precedents: corrected misinformation can continue to shape attitudes (Thorson 2016), language can stabilize social beliefs (Beukeboom and Burgers 2019), audience-tuned speech can feed back into later memory (Ye et al. 2021), and serial transmission can converge toward attractors (Kirby, Cornish, and Smith 2008). Finally, model-collapse work gives a training-time analogy for recursive degradation (Shumailov et al. 2023; Alemohammad et al. 2024; Seddik et al. 2024); we study a communication-time analogue that arises without retraining.

Measuring Social Fidelity

The probe is a controlled instrument, not a benchmark: we hold a society fixed, inject one ground-truthed change, and measure what survives transmission (Figure 1).

From reach to social fidelity. The task is designed to avoid a common ambiguity in multi-agent simulations. A message can reach the conversation stream without becoming the answer that agents later give. We therefore treat social fidelity as a three-stage measurement problem: **HEAR**, the value an agent is exposed to in its event stream; **SAY**, the value it later

utters in meetings; and **HOLD**, the value it gives when individually probed. The paper’s main claim is about the gap between **SAY** and **HOLD**, not about whether a string can be produced somewhere in the society.

Society and contact model. The society is $N=25$ LLM agents, each with a short persona describing a member of a neighborhood, and one designated source. In each round we draw m independent random matchings over the agents (duplicate pairs removed), so every agent holds about m pairwise conversations per round; we use $m=2$ unless noted, and vary $m \in \{1, 2, 3\}$ to test connectivity. Each meeting is a short exchange of T alternating turns ($T=3$), and both participants observe every utterance.

The ground-truthed update. The society is organizing an event with a known time and place—the stale value. At round 1 an authoritative update changes the event’s day and place to the current value, observed by the source. To guard against scenario-specific artifacts we use three structurally identical scenarios: **REPAIR_DRIVE** (Saturday, porch → Sunday, community center), **BOOK_CLUB** (Tuesday, library → Thursday, cafe), and **CARPOOL** (7 am, school lot → 8 am, church lot).

Propagation via reflection memory. Agents relay through generative-agent reflection memory (Park et al. 2023; Shinn et al. 2023; Zhong et al. 2024; Packer et al. 2024; Wang et al. 2023): an append-only stream of observed utterances plus periodic free-text reflections. Retrieval ranks the most relevant events and reflections (semantic embedding ranking with a lexical fallback) into the context each agent speaks and answers from, and agents consolidate (reflect) every round. This is the standard generative-agent baseline; we additionally test a currency-resolving memory variant that prioritizes the most recent value of a changing fact.

Interventions. The baseline injects the update once (round 1, into the source). An authoritative re-broadcast re-injects the update every round, either to the source only (which must then re-propagate it) or to every agent—a town-announcement / overwrite control. For the mechanism test we vary the timing of a single all-agent broadcast: early (round 1) versus late (immediately before the probe).

Measurement: speech versus held belief. **SAY** is computed from meeting logs as the fraction of value-coded utterances that carry the current value rather than the stale value. **HOLD** is measured separately by interviewing every agent after the run, instructing it to answer the probe (“when/where is it now?”) using only its own memory notes, and scoring the answer current, stale, or unknown by the scenario’s value markers. “Held belief” is operational—the answer an agent later gives when probed—and implies no claim about mental states. This operationalization is deliberately conservative: an agent that merely repeats the source in conversation receives no credit unless it later answers with the current value.

Models, replicates, and validation. The default agent model is gpt-5.4-mini at temperature 0; the capability ladder adds gpt-5.4 and gpt-5.5. Each condition is repeated over n



Figure 1: Social fidelity probe. We inject a ground-truthed update, observe social relay, and separately probe each agent’s later answer.

random seeds that control the meeting schedule (n up to 8). Because temperature 0 LLM societies are still chaotic (a single divergent token cascades), we report per-seed variation rather than treat any single run as canonical, and we validate the keyword verdict against an independent LLM judge that re-scores answers semantically (below), following the broader practice of behavioral factuality and hallucination evaluation (Lin, Hilton, and Evans 2022; Manakul, Liusie, and Gales 2023; Farquhar et al. 2024; Li et al. 2023c; Niu et al. 2024). Because elicited confidence and internal truth signals can diverge from surface text (Azaria and Mitchell 2023; Tian et al. 2023), we treat the final interview as an operational held-answer probe rather than a literal mind-read.

Design rationale. The probe uses a mutable fact rather than a static trivia item because social fidelity is most visible when an old value and a current value compete. The stale value is not noise: it is a previously valid coordination fact that agents have reason to reuse. The current value is ground-truthed and unambiguous, but it must travel through the same ordinary meetings and memory operations that make agent societies appealing as simulations. This creates a harder test than asking whether an isolated agent can answer a factual question, and a cleaner test than open-ended opinion diffusion, where there may be no external truth condition.

The intervention ladder is chosen to separate three explanations that are often conflated. If agents simply lack capability, stronger models or direct broadcast should help. If the problem is insufficient exposure, a persistent source should close the gap. If the problem is social state, only interventions that change many agents’ own stored value should work. The all-agent broadcast is therefore not treated as a realistic deployment mechanism; it is an upper-bound control that verifies the task is answerable and that the scoring rule can recognize success. The source condition is the stricter social test because it gives the truth an authoritative and repeated conversational origin while leaving adoption to the society.

Reporting convention. HOLD rates are percentages of agents at the final probe. SAY rates are computed over utterances that explicitly contain either the stale or current value; this prevents irrelevant conversation from being counted as false, but also means SAY is not a measure of how often the topic appears. A condition counts as a repair only if it moves final HOLD, not merely exposure or transcript frequency. This convention is why transient peaks in the long-horizon runs and source-driven increases in current-valued speech are treated as evidence for availability, not adoption.

Results

Speech is not belief

The central experiment compares three conditions (Figure 2). In the baseline, the update enters through ordinary social transmission. In the source condition, the authoritative source repeats the current truth across rounds. In broadcast, every agent receives the current truth directly.

The source intervention is the crucial test. It makes agents increasingly utter the current value: by the final round, 82% of value-coded source-condition utterances carry the current answer, compared with 63% in the baseline. Yet final interviews remain near baseline (15% held-current under source versus 12% under baseline, $n=5$). Broadcast to all moves both speech and belief (99% held-current), so the failure is not that the models cannot represent, retrieve, or output the current value. They can say it. The problem is that uttered truth does not become the answer agents later give when probed. A per-agent trace audit makes this stricter than the aggregate SAY curve: counting only utterances that contain both current markers (Sunday and community center), 28 source-condition agent-observations heard the full current pair from conversation, but 12 of them later answered unknown. Fifteen agent-observations themselves said the full current pair, yet four later answered unknown. The trace therefore rules out a purely transcript-level absence explana-

Speech vs. belief

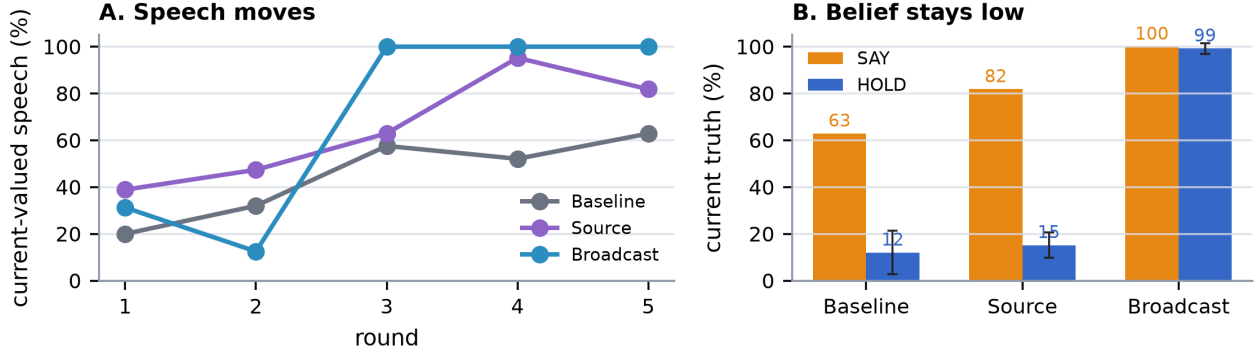


Figure 2: Speech is not belief. Source repetition moves speech, but only broad broadcast installs held truth. Error bars show 95% CIs.

tion. We use “held belief” only in this operational sense—the answer an agent gives from its own memory notes, not a claim about literal mental states.

Social fidelity decays over time

The current value reaches the conversation stream but does not reliably become the society’s held answer. Over a long horizon (Figure 3) truth can rise transiently—peaking near round 6—then decays toward a low steady state by round 30. The important object is the trajectory: truth can appear and still lose. A persistent source delays but does not prevent the erosion, while continuous broadcast keeps held-current belief near ceiling by bypassing ordinary social transmission.

The time series also changes the interpretation of the short-horizon experiments. Round 5 is near the transient peak, not a settled state: baseline held-current truth is about 20% at round 5, peaks at 28% around round 6, and falls to 6% by round 29. The source condition peaks higher early but falls to 4% by the end. The long-run failure is therefore not simply stale corruption; in many runs the update becomes unretrievable and the society becomes unknown-dominated. This strengthens the fidelity-decay claim: even when truth becomes visible for a time, ordinary social reuse does not keep it alive.

Natural levers do not repair held truth

Each lever is an intuitive repair story, and each fails (Figure 4). Capability helps modestly but leaves most agents off the current truth; connectivity is roughly neutral; a currency-resolving memory does not reliably rescue truth; a persistent authoritative source changes speech but not held belief; only broadcasting the truth to every agent every round works—an overwrite-style positive control, not a social cure. The claim is not that agents cannot output the truth; they can. Ordinary social communication does not install it as later held belief.

The negative result is important because the levers correspond to natural engineering responses. Stronger mod-

els raise held-current truth from 19% to roughly 30–34%, but most agents still fail the probe. Increasing meetings per round does not create redundancy-based error correction; at the tested power it is roughly neutral. The memory swap is similarly non-curative: it changes what gets relayed in some runs, but its held-current confidence interval overlaps the generative-agent baseline. Authority is the sharpest case because it does change the conversation stream. Its failure therefore rules out the simple story that agents merely need to hear a correction more often. This is where our data diverge from the optimistic reading of debate and multi-agent deliberation results: structured debate can improve judged answers (Du et al. 2024; Michael et al. 2023; Liang et al. 2024; Khan et al. 2024), but ordinary social relay can make the truth more utterable without making it durable.

The mechanism is early, broad entrenchment

If recency drove belief, a late all-agent broadcast just before the probe should work; if repetition drove it, a single early broadcast should fail. The opposite happens (Figure 5): early succeeds, late fails. The stale value begins with incumbent advantage; narrow or late correction can be uttered without taking over the society’s held state (Thorson 2016; Kirby, Cornish, and Smith 2008).

The mechanism requires both timing and breadth. Baseline is early but narrow: only the source receives the update, so held truth remains low. The persistent source is repeated but still narrow, and it also remains low. A late all-agent broadcast is broad but too late, landing near baseline despite being maximally recent. By contrast, a single early all-agent broadcast is enough to match every-round broadcast. The winning condition is not the number of times truth is repeated; it is whether the current value becomes the population-wide version before the stale incumbent has been reinforced by later conversation. Human communication work offers a useful precedent: linguistic framing and audience tuning can shape later memory and shared belief, rather than merely transmit-

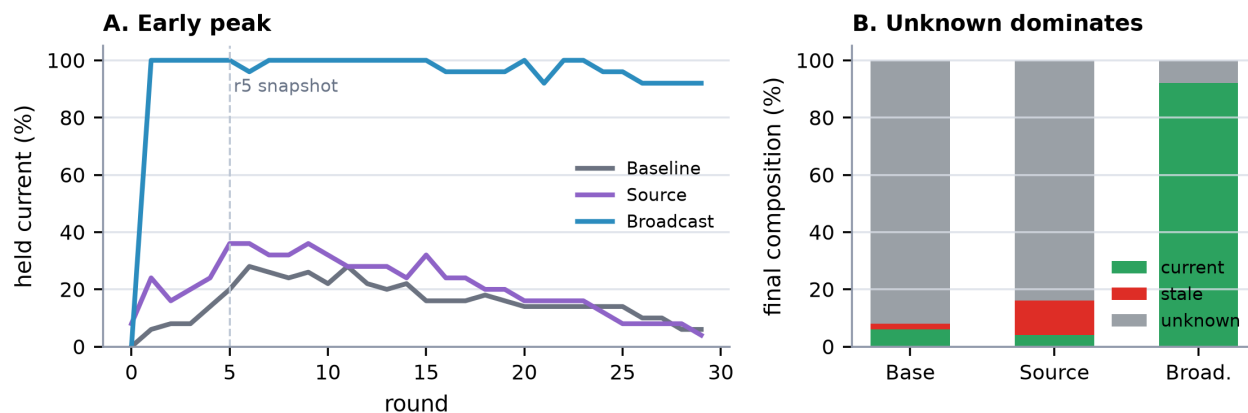


Figure 3: Long-horizon decay. Held truth peaks early, then erodes; continuous broadcast is the only stable condition.

ting neutral content (Beukeboom and Burgers 2019; Ye et al. 2021).

Robustness and measurement validation

The pattern replicates across REPAIR_DRIVE, BOOK_CLUB, and CARPOOL, and persists under thicker personas (Figure 6), addressing the concern that the effect is a thin-persona artifact (Park et al. 2024). A semantic LLM judge agrees with the keyword metric on the headline (99–100%), so the current/stale verdict is not a keyword artifact.

The scenario results also clarify the scope of the effect. Absolute difficulty varies: BOOK_CLUB has higher baseline retention than REPAIR_DRIVE or CARPOOL. But the intervention pattern is stable: source does not approach broadcast, while broadcast restores near-ceiling held truth in every scenario. The thick-persona condition addresses a different concern, that thin neighborhood sketches make agents too shallow to preserve facts. Richer persona descriptions leave the pattern unchanged. Finally, semantic judge validation shows that the main current/stale/unknown verdict is not an artifact of surface keyword matching.

Table 1 summarizes the full evidential chain. The important comparison is not any single percentage in isolation but the repeated separation between *local availability* and *population-wide adoption*. Source makes the correction available to the society and increasingly visible in conversation, yet the later private probe still recovers the stale or unknown state for most agents. Broadcast is therefore not a realistic repair proposal; it is a diagnostic upper bound showing that the agents and probe can represent the current value when the population state is forcibly aligned.

Data-level interpretation. The results separate three quantities that are easy to collapse: availability, adoption, and persistence. Source improves availability, visible as current-valued speech, but barely moves final private adoption. Broadcast moves adoption, ruling out a simple scoring or capability artifact. Long-horizon runs then show that adoption is not stable by default: early gains can decay into an unknown-dominated state. Scenario variation changes level, not pattern; we therefore read the small lifts from capability,

connectivity, or memory as partial availability effects, not repairs of held social state.

Discussion and Limitations

Implications. The central finding is a dissociation: in an LLM-agent society, what agents say and what the society holds can be moved independently. This cautions a growing practice of reading collective behavior off agent societies. An intervention that changes utterances need not change what the society later answers. It also sharpens the relation to single-agent fidelity: an agent can be faithful to an individual (Park et al. 2024) while a society of such agents fails to carry a fact. The failure is the communication-time cousin of recursive model collapse (Shumailov et al. 2023; Alemohammad et al. 2024): degradation from repeated social reuse, with no retraining.

For evaluation, this means that social-agent benchmarks should report at least three state variables rather than one: who heard the update, who later uttered it, and who privately answers with it after interaction. A transcript-only metric can mistake a surface correction for durable adoption, while a final-answer metric without transcript state can miss how the society arrived there.

Why the levers fail, and what would help. The mechanism is path-dependent entrenchment: the value established early and broadly is the one later conversation reinforces, so a late or narrow correction can be uttered without being adopted. This explains why each natural lever fails—more capable agents, more communication, or a different memory do not change which value arrived first—and why only overwriting every agent works. The constructive reading is that effective corrections must be early and broad by construction, and that memory architectures aimed at this failure must privilege the current value over the entrenched majority, not merely store more.

This suggests freshness-aware state reconciliation: for mutable facts, memory should treat later grounded updates as state replacements, mark older values obsolete, propagate that invalidation, and verify the private state after subse-

Natural levers fail

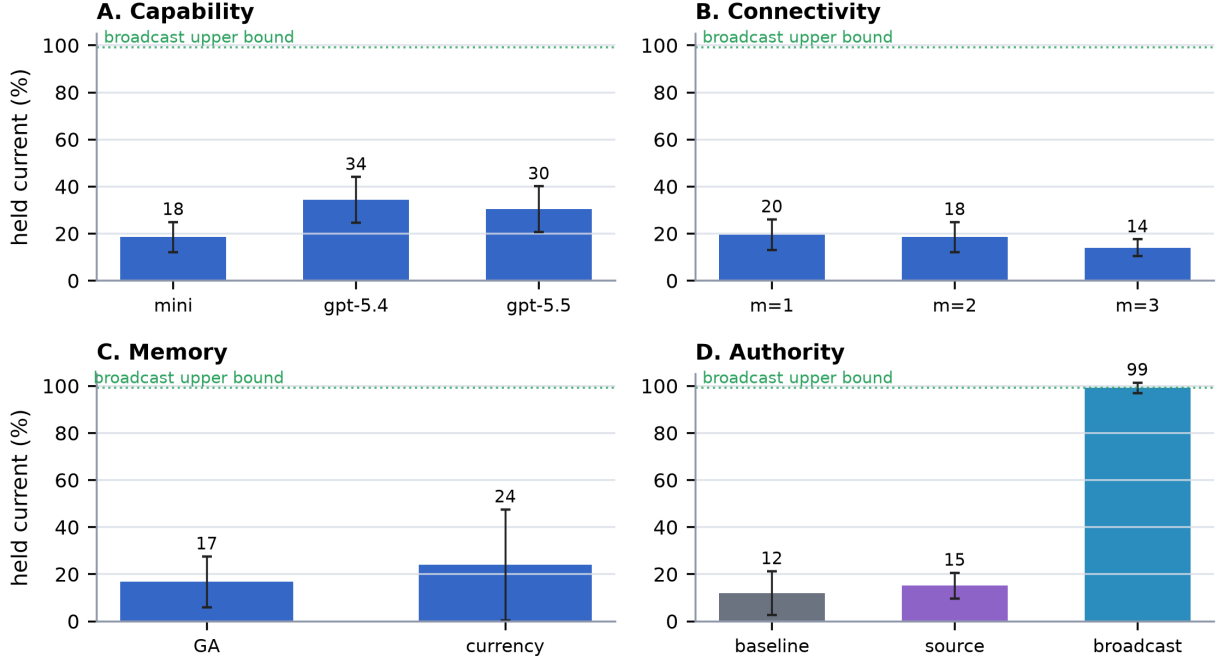


Figure 4: Natural levers fail. Capability, connectivity, memory, and source repetition all remain far below broadcast. Error bars show 95% CIs.

quent conversation. Adaptive memory systems such as A-MEM point toward this design space (Xu et al. 2025), but the present results show that memory capacity alone is not the missing ingredient.

Claim boundaries. We are deliberately conservative. We do not claim a capability or connectivity phase boundary; capability helps modestly and connectivity is roughly neutral in powered reruns (an early “connectivity amplifies corruption” cell was an underpowered outlier and did not replicate). We do not claim correction fails because agents never hear it—the source condition shows the opposite. And we do not claim memory never matters—only that the tested currency-resolving swap did not robustly restore held truth.

Limitations. The base task is a single family—an event whose day and place are rescheduled—and though it replicates across three scenarios, other fact types (quantities, causal or multi-part claims) are untested. The society is small ($N=25$) with a single source and a random-matching contact model; structured networks and larger populations remain open. “Held belief” is operationalized as the later probed answer, and both SAY and HOLD use value markers (judge-validated, but still surface-based). The unknown-dominated steady state may partly reflect reflection-memory retrieval; disentangling society-level from memory-level decay is future work. Results center on gpt-5.4-mini with a capability ladder to gpt-5.5; some conditions use modest n .

Future work. The next experiments should move from diagnosis to repair while preserving the HEAR/SAY/HOLD audit: explicit state invalidation, structured networks, multiple authorities, and other fact types.

Conclusion

In LLM-agent societies, a society can learn to say the truth without coming to hold it. The dissociation is driven by path-dependent entrenchment, survives a long horizon and several robustness checks, and resists every natural repair lever short of overwriting memory directly. Social-agent reliability therefore cannot be read from transcript fluency alone: a correction has done real work only when it changes the answer agents give after later interaction.

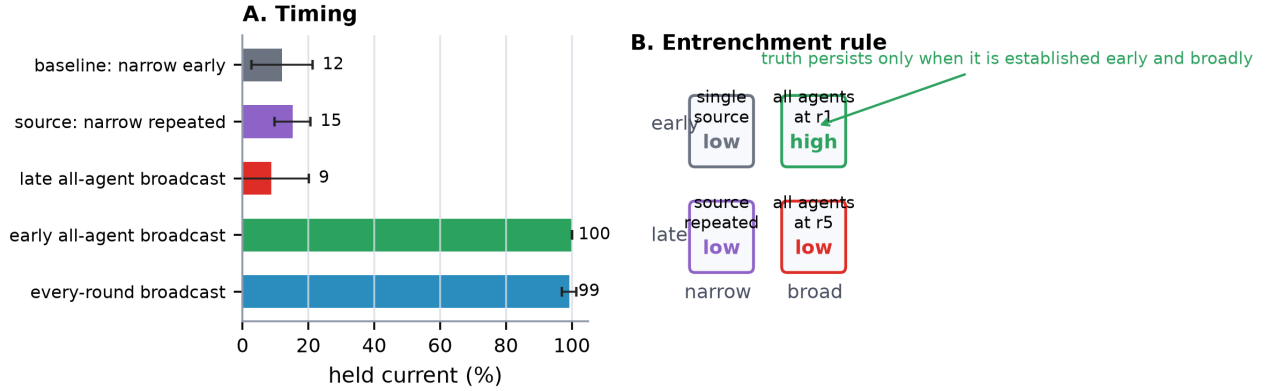


Figure 5: Early broad entrenchment. Early broadcast succeeds; late broadcast fails despite maximal recency.

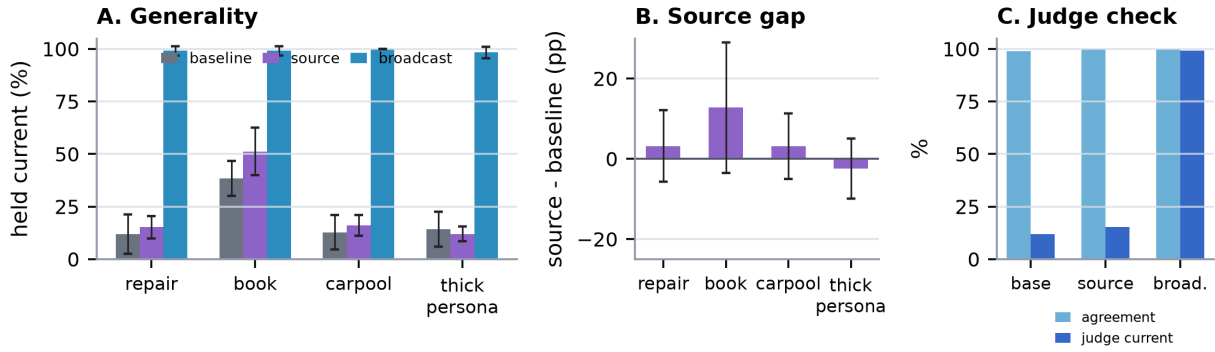


Figure 6: Robustness and validation. The source–broadcast gap holds across scenarios, thicker personas, and semantic re-scoring.

Evidence block	Question	Result	Supported claim
Speech–belief probe	Does an authority source repair held belief?	Source raises final SAY-current to 82%, but HOLD-current stays near baseline (15% vs. 12%); broadcast reaches 99%.	Speech and belief are separable.
Long horizon	Is the truth merely slow to settle?	Baseline peaks at 28% and decays to 6%; source peaks at 36% and decays to 4%; broadcast remains near ceiling.	Social truth decays without broad overwrite.
Natural levers	Do scale-like repairs fix the problem?	Capability reaches only 30–34%, connectivity is neutral, memory is unstable, and authority remains near baseline.	Common engineering levers are not sufficient.
Timing and breadth	Is the mechanism recency or repetition?	Late all-agent broadcast fails (9%); one early all-agent broadcast succeeds (100%).	Early, broad entrenchment governs adoption.
Robustness checks	Is the effect a scenario, persona, or metric artifact?	The source–broadcast gap holds across scenarios and thick personas; semantic judge scoring reproduces the headline metric.	The pattern is not a thin setup artifact.

Table 1: Evidence ledger. Each block isolates a different link in the social-fidelity failure chain.

References

- Alemohammad, S.; Casco-Rodriguez, J.; Luzi, L.; Humayun, A. I.; Babaei, H.; LeJeune, D.; Siahkoohi, A.; and Baraniuk, R. G. 2024. Self-Consuming Generative Models Go MAD. In *International Conference on Learning Representations (ICLR)*.
- Azaria, A.; and Mitchell, T. 2023. The Internal State of an LLM Knows When It’s Lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 967–976.
- Beukeboom, C. J.; and Burgers, C. 2019. How Stereotypes Are Shared Through Language: A Review and Introduction of the Social Categories and Stereotypes Communication Framework. *Review of Communication Research*, 7: 1–37.
- Chen, W.; Su, Y.; Zuo, J.; Yang, C.; Yuan, C.; Chan, C.-M.; Yu, H.; Lu, Y.; Hung, Y.-H.; Qian, C.; Qin, Y.; Cong, X.; Xie, R.; Liu, Z.; Sun, M.; and Zhou, J. 2023. AgentVerse: Facilitating Multi-Agent Collaboration and Exploring Emergent Behaviors in Agents. *arXiv preprint arXiv:2308.10848*.
- Du, Y.; Li, S.; Torralba, A.; Tenenbaum, J. B.; and Mordatch, I. 2024. Improving Factuality and Reasoning in Language Models through Multiagent Debate. In *International Conference on Machine Learning (ICML)*.
- Farquhar, S.; Kossen, J.; Kuhn, L.; and Gal, Y. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630: 625–630.
- Friggeri, A.; Adamic, L. A.; Eckles, D.; and Cheng, J. 2014. Rumor Cascades. In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*.
- Gao, D.; Li, Z.; Pan, X.; Kuang, W.; Ma, Z.; Qian, B.; Wei, F.; Zhang, W.; Xie, Y.; Chen, D.; et al. 2024. AgentScope: A Flexible yet Robust Multi-Agent Platform. *arXiv preprint arXiv:2402.14034*.
- Hong, S.; Zhuge, M.; Chen, J.; Zheng, X.; Cheng, Y.; Zhang, C.; Wang, J.; Wang, Z.; Yau, S. K. S.; Lin, Z.; Zhou, L.; Ran, C.; Xiao, L.; Wu, C.; and Schmidhuber, J. 2024. MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework. In *International Conference on Learning Representations (ICLR)*.
- Huang, J.; Chen, X.; Mishra, S.; Zheng, H. S.; Yu, A. W.; Song, X.; and Zhou, D. 2024. Large Language Models Cannot Self-Correct Reasoning Yet. In *International Conference on Learning Representations (ICLR)*.
- Kadavath, S.; Conerly, T.; Askell, A.; Henighan, T.; Drain, D.; Perez, E.; Schiefer, N.; Hatfield-Dodds, Z.; DasSarma, N.; Tran-Johnson, E.; et al. 2022. Language Models (Mostly) Know What They Know. *arXiv preprint arXiv:2207.05221*.
- Khan, A.; Hughes, J.; Valentine, D.; Ruis, L.; Sachan, K.; Radhakrishnan, A.; Grefenstette, E.; Bowman, S. R.; Rocktaschel, T.; and Perez, E. 2024. Debating with More Persuasive LLMs Leads to More Truthful Answers. In *International Conference on Machine Learning (ICML)*.
- Kirby, S.; Cornish, H.; and Smith, K. 2008. Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *Proceedings of the National Academy of Sciences*, 105(31): 10681–10686.
- Lazer, D. M. J.; Baum, M. A.; Benkler, Y.; Berinsky, A. J.; Greenhill, K. M.; Menczer, F.; Metzger, M. J.; Nyhan, B.; Pennycook, G.; Rothschild, D.; Schudson, M.; Sloman, S. A.; Sunstein, C. R.; Thorson, E. A.; Watts, D. J.; and Zittrain, J. L. 2018. The science of fake news. *Science*, 359(6380): 1094–1096.
- Li, G.; Hammoud, H. A. A. K.; Itani, H.; Khizbullin, D.; and Ghanem, B. 2023a. CAMEL: Communicative Agents for “Mind” Exploration of Large Language Model Society. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Li, H.; Chong, Y. Q.; Stepputtis, S.; Campbell, J.; Hughes, D.; Lewis, M.; and Sycara, K. 2023b. Theory of Mind for Multi-Agent Collaboration via Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Li, J.; Cheng, X.; Zhao, W. X.; Nie, J.-Y.; and Wen, J.-R. 2023c. HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Liang, T.; He, Z.; Jiao, W.; Wang, X.; Wang, Y.; Wang, R.; Yang, Y.; Shi, S.; and Tu, Z. 2024. Encouraging Divergent Thinking in Large Language Models through Multi-Agent

- Debate. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Lin, S.; Hilton, J.; and Evans, O. 2022. TruthfulQA: Measuring How Models Mimic Human Falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Manakul, P.; Liusie, A.; and Gales, M. J. F. 2023. SelfCheck-GPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Maurya, R. G.; Shukla, V.; Dandekar, R. A.; Dandekar, R.; and Panat, S. 2025. Simulating Misinformation Propagation in Social Networks using Large Language Models. *arXiv preprint arXiv:2511.10384*.
- Michael, J.; Mahdi, S.; Rein, D.; Petty, J.; Dirani, J.; Padmakumar, V.; and Bowman, S. R. 2023. Debate Helps Supervise Unreliable Experts. *arXiv preprint arXiv:2311.08702*.
- Niu, C.; Wu, Y.; Zhu, J.; Xu, S.; Shum, K.; Zhong, R.; Song, J.; and Zhang, T. 2024. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Packer, C.; Wooders, S.; Lin, K.; Fang, V.; Patil, S. G.; Stoica, I.; and Gonzalez, J. E. 2024. MemGPT: Towards LLMs as Operating Systems. In *International Conference on Learning Representations (ICLR)*.
- Park, J. S.; O'Brien, J. C.; Cai, C. J.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.
- Park, J. S.; Zou, C. Q.; Kamphorst, J.; Egan, N.; Shaw, A.; Hill, B. M.; Cai, C.; Morris, M. R.; Liang, P.; Willer, R.; and Bernstein, M. S. 2024. Generative Agent Simulations of 1,000 People. *arXiv preprint arXiv:2411.10109*.
- Qian, C.; Liu, W.; Liu, H.; Chen, N.; Dang, Y.; Li, J.; Yang, C.; Chen, W.; Su, Y.; Cong, X.; Xu, J.; Li, D.; Liu, Z.; and Sun, M. 2024. ChatDev: Communicative Agents for Software Development. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 15174–15186.
- Seddik, M. E. A.; Youssef, P.; Chen, S.-W.; Hayou, S.; and Debbah, M. 2024. How Bad is Training on Synthetic Data? A Statistical Analysis of Language Model Collapse. *arXiv preprint arXiv:2404.05090*.
- Shinn, N.; Cassano, F.; Berman, E.; Gopinath, A.; Narasimhan, K.; and Yao, S. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Shumailov, I.; Shumaylov, Z.; Zhao, Y.; Gal, Y.; Papernot, N.; and Anderson, R. 2023. The Curse of Recursion: Training on Generated Data Makes Models Forget. *arXiv preprint arXiv:2305.17493*.
- Thorson, E. 2016. Belief Echoes: The Persistent Effects of Corrected Misinformation. *Political Communication*, 33(3): 460–480.
- Tian, K.; Mitchell, E.; Zhou, A.; Sharma, A.; Rafailov, R.; Yao, H.; Finn, C.; and Manning, C. D. 2023. Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 5433–5442.
- Vosoughi, S.; Roy, D.; and Aral, S. 2018. The spread of true and false news online. *Science*, 359(6380): 1146–1151.
- Wang, G.; Xie, Y.; Jiang, Y.; Mandlekar, A.; Xiao, C.; Zhu, Y.; Fan, L.; and Anandkumar, A. 2023. Voyager: An Open-Ended Embodied Agent with Large Language Models. *Transactions on Machine Learning Research*.
- Wu, Q.; Bansal, G.; Zhang, J.; Wu, Y.; Li, B.; Zhu, E.; Jiang, L.; Zhang, X.; Zhang, S.; Liu, J.; Awadallah, A. H.; White, R. W.; Burger, D.; and Wang, C. 2023. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. *arXiv preprint arXiv:2308.08155*.
- Xu, W.; Liang, Z.; Mei, K.; Gao, H.; Tan, J.; and Zhang, Y. 2025. A-Mem: Agentic Memory for LLM Agents. *arXiv preprint arXiv:2502.12110*.
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR)*.
- Ye, J.; Zhao, L.; Huang, Z.; and Meng, F. 2021. The Audience-Tuning Effect of Negative Stereotypes in Communication. *Frontiers in Psychology*, 12: 663814.
- Zhong, W.; Guo, L.; Gao, Q.; Ye, H.; and Wang, Y. 2024. MemoryBank: Enhancing Large Language Models with Long-Term Memory. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(17): 19724–19731.
- Zhou, X.; Zhu, H.; Mathur, L.; Zhang, R.; Yu, H.; Qi, Z.; Morency, L.-P.; Bisk, Y.; Fried, D.; Neubig, G.; and Sap, M. 2024. SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents. In *International Conference on Learning Representations (ICLR)*.